

DEVOPS MASTER COURSE

Volume 8B

Google Cloud Core Infrastructure

Chapters 11-20

Compute Fleets • Managed Instance Groups • Shared VPC • Security • Load Balancing • DNS •
Storage • Hybrid Connectivity • Backup • Disaster Recovery

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About This Part

Volume 8B moves from single-resource foundations to resilient Google Cloud infrastructure. The focus is not only on creating virtual machines, networks, and storage. The focus is on building replaceable fleets, centrally governed connectivity, health-aware application delivery, protected data, recoverable operations, and evidence-driven production practices.

Each chapter explains design intent before implementation. Labs emphasize least privilege, private connectivity, rollback, recovery testing, cost cleanup, and the difference between availability, backup, and disaster recovery.

Volume 8 Progress

Part	Coverage	Status
8A	Foundations: hierarchy, IAM, tools, compute, VPC, billing	Complete
8B	Core infrastructure: fleets, networking, load balancing, DNS, storage, connectivity, DR	This book
8C	Databases, analytics, data engineering, AI-ready data platforms, migration	Next
8D	GKE, Cloud Run, functions, APIs, Pub/Sub, Eventarc, workflows	Planned
8E	Terraform, CI/CD, observability, security, governance, FinOps	Planned
8F	Enterprise projects, incidents, certifications, interviews, final capstone	Planned

Safety, Security, and Cost Controls

- Use a dedicated learning project and explicit gcloud configuration. Verify project, region, zone, and active principal before every change.
- Create budgets and cleanup plans before deploying load balancers, static addresses, VPN gateways, Interconnect attachments, backup vaults, or large storage resources.
- Prefer private VMs, Identity-Aware Proxy, OS Login, service account impersonation, and narrowly scoped runtime service accounts. Avoid public SSH and long-lived keys.
- Test firewall, route, DNS, load-balancer, and recovery behavior with allowed and denied evidence rather than relying on console configuration alone.
- Use lifecycle rules carefully. Retention locks and immutable backup settings can be irreversible and billable.
- Delete temporary forwarding rules, backend services, instance groups, disks, snapshots, buckets, IP addresses, and hybrid resources after labs.

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Chapter 11 - Advanced Compute Engine and Fleet Operations

Engineering virtual machines as governed, observable, replaceable infrastructure

Learning Objectives

- Design VM configurations for security, performance, availability, and lifecycle management.
- Use instance templates, custom images, startup automation, Shielded VM, and confidential-computing options appropriately.
- Explain maintenance behavior, Local SSD trade-offs, reservations, and spot capacity.
- Operate VM fleets through policy, patching, logging, and controlled access.

11.1 From Individual VMs to Managed Fleet Standards

A production VM should be created from an approved configuration rather than assembled manually. The standard includes machine family, image source, disk type, network tags or secure tags, service account, Shielded VM settings, metadata, startup behavior, logging, backup policy, maintenance behavior, and ownership labels.

- Treat the boot disk as replaceable; store durable application data on managed data services or protected data disks.
- Use image families or versioned custom images with tested rollback.
- Use instance templates for repeatable fleet creation.
- Record business owner, environment, application, data classification, and expiry through labels or tags.

11.2 Machine Families, Scheduling, and Capacity

Machine selection is a workload decision, not a default-size decision. General-purpose families suit broad workloads; compute-, memory-, storage-, accelerator-, and specialized families target specific bottlenecks. Capacity must also be considered by region and zone. Reservations can protect required capacity, while Spot VMs reduce cost for interruption-tolerant workloads.

Requirement	Preferred Direction	Operational Guardrail
Steady general workload	General-purpose machine family	Measure CPU, memory, disk, and network before rightsizing.
Burst or batch	Autoscaled fleet or Spot VMs	Checkpoint work and tolerate termination.
Guaranteed launch capacity	Reservation or capacity planning	Track utilization and expiration.
Sensitive in-use data	Confidential VM where supported	Validate image, machine, and performance compatibility.
Very low latency scratch	Local SSD	Design for data loss and host-event behavior.

11.3 Identity, Metadata, and Administrative Access

Attach a dedicated service account with the minimum roles needed by the workload. Avoid broad OAuth scopes and service-account keys. Use OS Login and Identity-Aware Proxy TCP forwarding for controlled administration when possible. Metadata and startup scripts are powerful configuration channels and must not contain plaintext secrets.

```
gcloud compute instances describe VM_NAME --zone=ZONE
gcloud compute instances get-serial-port-output VM_NAME --zone=ZONE
gcloud compute ssh VM_NAME --zone=ZONE --tunnel-through-iap
```

11.4 Maintenance, Patching, and Observability

Compute Engine can live-migrate many VMs during infrastructure maintenance, but some accelerators, specialized machine types, or Local SSD configurations can require termination behavior. Define automatic restart, maintenance policy, patch windows, and recovery expectations. Install the Ops Agent where appropriate and centralize logs, metrics, audit evidence, and inventory.

- Use VM Manager or controlled image replacement for patch compliance.
- Monitor guest health, disk pressure, startup failures, and service-account permission errors.
- Prefer immutable replacement over repeated manual repair.
- Validate deletion protection and disk auto-delete behavior before decommissioning.

11.5 Image and Configuration Promotion

A controlled image pipeline creates a candidate image, scans and tests it, deploys it to a nonproduction fleet, captures health evidence, and promotes an immutable version. Emergency patching must still converge back to the approved configuration; otherwise fleets accumulate drift.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
VM starts but application is unavailable	Startup script, guest service, firewall, or dependency failure	Inspect serial output, startup-script logs, guest service status, firewall logging, and dependency DNS/routes.
SSH is denied	OS Login/IAP/IAM/firewall path is incomplete	Verify IAP role, OS Login role, TCP 22 from IAP range if required, guest configuration, and audit logs.
VM cannot call Google APIs	Wrong service account, IAM, scope, or private access	Inspect attached identity, IAM policy, access scopes, Private Google Access/NAT, and API enablement.
Unexpected VM termination	Spot preemption, maintenance policy, quota, or host event	Review instance status, operations, maintenance logs, Spot design, and restart policy.

Hands-on Lab 11: Build a Secure Versioned VM Standard

Prerequisites

- A dedicated project, custom VPC/subnet, and permissions to create Compute Engine resources.

Implementation

1. Create a dedicated runtime service account with one read-only role needed by the demo workload.
2. Create a startup script that installs a small web service without embedding secrets.
3. Create an instance template using a supported image family, private IP only, Shielded VM features, labels, and the dedicated service account.
4. Launch a test VM from the template and access it through IAP or another approved private path.
5. Capture serial output, guest logs, service-account identity, disk settings, and allowed/denied API tests.

6. Create a second template version and document rollback to the first version.

Validation Evidence

- The VM has no public IP.
- The runtime identity cannot perform an unrelated privileged action.
- The application survives a controlled restart.
- Template versions and rollback instructions are recorded.

Cleanup

- Delete the test VM and unused template versions. Verify that no orphaned disks, reservations, or static addresses remain.

Interview Questions and Model Answers

1. Why use instance templates for VM fleets?

They provide versioned, repeatable configuration for managed instance groups and reduce manual drift.

2. When is Local SSD appropriate?

For high-performance temporary data that can be lost and reconstructed when the VM or host fails.

3. Why prefer IAP and OS Login over public SSH?

They reduce public exposure and centralize identity, authorization, and audit evidence.

4. What is immutable VM operations?

Updating a fleet by replacing instances from a tested image or template rather than repeatedly modifying long-lived servers.

Chapter Summary

- Production VMs require approved identity, image, network, disk, and maintenance standards.
- Private administration and short-lived identity reduce exposure.
- Local SSD and Spot capacity require interruption-aware design.
- Replacement-based operations limit drift and improve rollback.

Chapter 12 - Managed Instance Groups, Autoscaling, and Safe Deployments

Building self-healing, multi-zone Compute Engine fleets with controlled change

Learning Objectives

- Differentiate managed and unmanaged instance groups.
- Design regional managed instance groups with health checks and autohealing.
- Select autoscaling signals, initialization periods, and scale-in controls.
- Use rolling and canary updates with rollback evidence.

12.1 Managed Instance Group Architecture

A managed instance group (MIG) creates and maintains VMs from an instance template. It can autoheal unhealthy instances, autoscale, distribute instances across zones, and perform rolling or canary updates. Regional MIGs improve resilience to zonal failure by distributing capacity across multiple zones in a region.

Group Type	Primary Use	Key Limitation
Zonal MIG	Simple zonal fleet or queue worker	A zone outage can affect the whole group.
Regional MIG	Highly available regional application	Requires regional capacity planning and balanced distribution.
Stateful MIG	VM identity or selected disks/IPs must persist	More complex recovery and scaling semantics.
Unmanaged group	Existing heterogeneous VMs	No template-driven autohealing or autoscaling.

12.2 Health Checks and Autohealing

Load-balancer health checks decide whether traffic should reach a backend. MIG autohealing health checks decide whether a VM should be repaired or recreated. These functions can use similar probes but have different consequences. An overly aggressive autohealing policy can create a repair storm during dependency outages.

- Probe application readiness, not only whether the operating system is running.
- Set an initial delay that covers startup and dependency initialization.
- Separate transient dependency failure from true instance corruption.
- Log health-state changes and repair actions.

12.3 Autoscaling as a Feedback System

Autoscaling can use CPU utilization, load-balancing capacity, Cloud Monitoring metrics, schedules, or queue-based signals such as Pub/Sub for supported zonal scenarios. Stable scaling requires a signal aligned with user demand, realistic initialization time, maximum capacity, and downstream limits.

Signal	Good Fit	Risk
CPU utilization	CPU-bound stateless services	Weak for I/O or latency-bound applications.

Signal	Good Fit	Risk
Load-balancer capacity	Request-serving web tiers	Requires correct backend capacity settings.
Monitoring metric	Business or application-specific demand	Bad metric design can oscillate or lag.
Schedule	Known recurring load	Cannot react alone to unexpected demand.
Queue signal	Asynchronous workers	Must coordinate visibility timeout, retries, and idempotency.

12.4 Rolling, Canary, and Opportunistic Updates

MIG updates change the intended configuration and apply it according to an update policy. Proactive rolling updates replace instances automatically. Canary updates expose a subset of the fleet to a new template. Opportunistic policies update instances when they are otherwise recreated. Configure maximum surge and maximum unavailable values based on capacity and risk.

```
gcloud compute instance-groups managed rolling-action start-update MIG_NAME \
--version=template=NEW_TEMPLATE \
--max-surge=1 --max-unavailable=0 --region=REGION
```

12.5 Stateful MIGs and Data Boundaries

Stateful MIGs preserve selected disks, internal IP addresses, metadata, or instance names through recreation. They can improve operations for some stateful systems, but they do not remove the need for application replication, backup, fencing, and recovery testing. Use managed databases where they better match the workload.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
MIG continuously recreates instances	Health check is too strict or startup delay is too short	Review autohealing logs, application startup time, dependency health, and initial delay.
Autoscaler oscillates	Signal noise, short initialization, or conflicting policies	Graph signal and capacity, increase stabilization, select one demand-aligned metric, and validate downstream limits.
Update stalls	Insufficient quota/capacity or restrictive surge/unavailable values	Inspect group operations, quota, zone capacity, template validity, and update policy.
Canary is healthy but broad rollout fails	Canary sample or traffic was not representative	Use weighted traffic, longer observation, representative zones, and automated rollback criteria.

Hands-on Lab 12: Deploy a Regional Self-Healing Web Fleet

Prerequisites

- A versioned instance template and two or more zones in one region.

Implementation

1. Create an application health check and validate it directly.
2. Create a regional MIG with a minimum of two instances distributed across zones.
3. Configure autohealing with a realistic initial delay.
4. Attach autoscaling based on load-balancing capacity or a suitable monitoring metric.
5. Deploy a second template as a canary to a small percentage of instances.
6. Generate load, observe scaling, intentionally fail one instance, and record repair behavior.
7. Promote or roll back the canary using documented thresholds.

Validation Evidence

- Traffic remains available during one instance failure.
- The MIG repairs the unhealthy instance.
- Autoscaling changes capacity without exceeding the configured maximum.
- Canary health and rollback evidence are preserved.

Cleanup

- Remove autoscaling and delete the MIG, templates, and health-check resources that are not required for later labs.

Interview Questions and Model Answers

1. What is the difference between load-balancer health and autohealing health?

Load-balancer health controls traffic routing; autohealing health can trigger VM recreation.

2. Why use a regional MIG?

To distribute a fleet across zones and reduce the impact of a zonal outage.

3. What causes autoscaling thrash?

Noisy or lagging signals, unrealistic initialization, conflicting policies, and a narrow capacity range.

4. What should govern a rolling update?

Health evidence, surge/unavailable capacity, representative canaries, rollback criteria, quota, and downstream capacity.

Chapter Summary

- MIGs deliver template-driven repair, scaling, and deployment.
- Regional groups protect stateless fleets from zonal failure.
- Health checks and initialization settings must reflect real application behavior.
- Canary and rollback evidence are mandatory for safe fleet change.

Chapter 13 - Enterprise VPC and Shared VPC Architecture

Centralizing network governance while preserving workload autonomy

Learning Objectives

- Design multi-project network boundaries using Shared VPC.
- Explain host projects, service projects, subnet IAM, and delegated administration.
- Plan IP ranges, routes, peering, Private Service Connect, and private services access.
- Operate global VPC networks with regional failure and ownership awareness.

13.1 Global VPC, Regional Subnets, and Project Boundaries

A Google Cloud VPC network is global, while its subnets are regional. This model allows one network to span regions, but address planning, routing, firewall policy, and service availability still require regional design. Projects are administrative boundaries; Shared VPC allows a centrally governed network in a host project to be consumed by service projects.

Component	Scope	Ownership Consideration
VPC network	Global	Usually central network or workload team.
Subnet	Regional	Delegated through subnet-level IAM when needed.
Route	Network-level with regional/global behavior by type	Central route governance prevents accidental transitive paths.
Firewall policy	Organization/folder/network hierarchy	Security administration should be separated from workload deployment.
Service project resource	Project and region/zone	Workload team owns lifecycle; network interface uses host-project subnet.

13.2 Shared VPC Operating Model

A host project owns the Shared VPC network, subnets, routes, and network-level controls. Service projects create resources such as VMs and GKE clusters that attach to authorized host-project subnets. Shared VPC administrators, network administrators, security administrators, and service-project administrators should be separate roles.

- Grant Network User at the narrowest practical subnet scope.
- Create Cloud NAT and shared network services in the host project.
- Centralize DNS and firewall policy while preserving workload project IAM.
- Use organization policy to restrict unauthorized Shared VPC patterns.

13.3 IP Address Management and Route Design

IP planning must account for primary ranges, secondary ranges, hybrid networks, service producer ranges, future regions, mergers, and multicloud connections. Overlapping ranges block routing options and complicate migration. Dynamic routes learned through Cloud Router are controlled by routing mode and BGP policy.

- Reserve contiguous non-overlapping ranges by region and environment.
- Document route ownership, next hops, priority, and propagation.
- Use Connectivity Tests and route inspection before changing production paths.

- Avoid assuming VPC Network Peering is transitive; it is not.

13.4 Private Service Connectivity

Private Google Access lets private workloads reach supported Google APIs without external IP addresses. Private services access allocates address space and creates private connectivity to service-producer networks for supported managed services. Private Service Connect exposes producer or Google services through private endpoints or published services with clearer consumer-producer boundaries.

Mechanism	Use	Key Design Point
Private Google Access	Private VM to Google APIs	Enabled per subnet and depends on DNS/routing design.
Private services access	VPC to supported managed producer network	Allocate non-overlapping producer range in the correct project.
Private Service Connect endpoint	Consumer private endpoint to service	Use endpoint IAM, DNS, and service-attachment controls.
VPC Peering	Private connectivity between VPCs	Non-transitive; quotas and route exchange require planning.

13.5 Enterprise Network Topologies

Common topologies include centralized Shared VPC, multiple environment-specific Shared VPCs, hub-and-spoke with Network Connectivity Center, and isolated application VPCs connected through approved services. The right model balances blast radius, autonomy, route scale, security ownership, and migration complexity.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
Service project cannot attach to subnet	Missing Shared VPC association or subnet Network User permission	Verify host/service attachment, subnet IAM, service agent permissions, and organization policy.
Private managed service cannot allocate address	Producer range missing, overlapping, or created in wrong project	Inspect allocated ranges and create private service connection in the host project for Shared VPC.
Networks connect but routes are missing	Peering export/import, dynamic routing mode, BGP, or non-transitive assumption	Inspect effective routes, peering settings, Cloud Router advertisements, and topology.
Address plan blocks expansion	Ranges overlap with on-premises or future region	Create an IPAM source of truth and redesign before adding dependencies.

Hands-on Lab 13: Design and Validate a Shared VPC Model

Prerequisites

- Organization access is optional; the lab can be completed as a detailed design if Shared VPC permissions are unavailable.

Implementation

1. Define host and service project responsibilities for network, security, DNS, billing, and workload teams.
2. Create a regional IP allocation plan for production, nonproduction, services, and hybrid connectivity.
3. Create or model a custom Shared VPC host network with at least two regional subnets.
4. Delegate one subnet to a service-project administrator at subnet scope.
5. Enable private Google API access and document private service connectivity requirements.
6. Run route and connectivity validation for workload-to-workload, workload-to-Google API, and denied paths.

Validation Evidence

- The service project can use only its approved subnet.
- Address ranges do not overlap with documented external networks.
- Central network and workload permissions are separated.
- Allowed and denied paths have evidence.

Cleanup

- Detach and delete temporary service projects or networks only after confirming no dependent resources remain.

Interview Questions and Model Answers**1. What does the Shared VPC host project own?**

The shared network, subnets, routes, and central network controls consumed by attached service projects.

2. Is a Google Cloud VPC regional?

No. The VPC is global; subnets are regional.

3. Is VPC Network Peering transitive?

No. A network cannot automatically route through one peer to reach another peer.

4. When use Private Service Connect?

When consumers need private endpoints to Google, third-party, or internally published services with controlled producer-consumer boundaries.

Chapter Summary

- Shared VPC centralizes network governance across projects.
- Subnet IAM supports delegated administration.
- IP and route design must anticipate hybrid and managed-service ranges.
- Private connectivity mechanisms solve different service-access problems.

Chapter 14 - Network Security, Routing, NAT, and Secure Administration

Applying layered controls to ingress, egress, lateral movement, and administration

Learning Objectives

- Use hierarchical firewall policies, global/regional network firewall policies, and VPC firewall rules correctly.
- Design route priority, Cloud Router, and NAT behavior.
- Implement private administration through IAP, OS Login, and controlled bastion patterns.
- Troubleshoot network paths using logs and Connectivity Tests.

14.1 Firewall Evaluation and Policy Hierarchy

Google Cloud supports hierarchical firewall policies at organization or folder scope, network firewall policies, and VPC firewall rules. Policies can enforce central guardrails while workload teams manage narrower rules. Both allowed and denied traffic should be observable through logging where the risk justifies the cost.

Control	Scope	Best Use
Hierarchical firewall policy	Organization or folder	Enterprise guardrails and delegated policy.
Global network firewall policy	One or more VPC networks	Reusable central network rules.
Regional network firewall policy	Regional resources/traffic where supported	Regional enforcement and segmentation.
VPC firewall rule	Single VPC network	Workload-specific ingress and egress.
Cloud Armor policy	Application/network edge behind supported load balancers	DDoS and application-layer protection.

14.2 Targets, Identities, and Least Privilege

Firewall rules can target resources through network tags or secure tags and, for supported cases, service accounts. Tags must be governed because changing a tag can change network access. Use narrow source ranges, protocol/port sets, and destination controls. Egress is part of the security boundary, not an afterthought.

14.3 Routes, Cloud Router, and Dynamic Exchange

Routes select next hops based on destination prefix, priority, applicability, and route type. Cloud Router exchanges dynamic routes by BGP for HA VPN and Interconnect; it does not forward packets itself. Custom route advertisements and routing mode determine which prefixes are exchanged.

- Document route priority and ownership.
- Avoid broad 0.0.0.0/0 custom routes without controlled next hops.
- Use Network Intelligence Center and Connectivity Tests before and after route changes.
- Monitor BGP session state and learned/advertised prefixes.

14.4 Public and Private Cloud NAT

Public Cloud NAT provides outbound internet connectivity for supported private resources without assigning external IP addresses. It is regional and depends on Cloud Router configuration. Private NAT translates

addresses between private networks for supported scenarios. NAT does not replace firewall policy and does not allow unsolicited inbound connections.

```
gcloud compute routers nats describe NAT_NAME \
  --router=ROUTER_NAME --region=REGION
gcloud compute routers get-status ROUTER_NAME --region=REGION
```

14.5 Secure Administration and Troubleshooting

Use IAP TCP forwarding with OS Login for controlled VM administration where feasible. For broader enterprise access, combine identity-aware administration with private connectivity, just-in-time privilege, logging, and session controls. Packet Mirroring, VPC Flow Logs, firewall logs, Cloud NAT logs, and Connectivity Tests provide complementary evidence.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
Private VM has no internet egress	Cloud NAT, route, firewall, DNS, or subnet selection problem	Check default/custom routes, NAT subnet coverage, source ports, egress rules, DNS, and NAT logs.
Firewall allow rule seems ignored	Higher-priority deny or wrong target/source	Inspect effective firewall policies across hierarchy, priorities, tags/service accounts, and logging.
BGP session is down	ASN, peer IP, tunnel/attachment, or MD5/config mismatch	Check Cloud Router status, peer device, HA VPN/Interconnect health, and advertised routes.
IAP SSH fails	IAM, OS Login, firewall, guest agent, or IAP range issue	Validate IAP tunnel role, OS Login role, ingress rule, guest agent, and serial logs.

Hands-on Lab 14: Create a Private Egress and Administration Pattern

Prerequisites

- A custom VPC with a private subnet and permissions for Cloud Router, Cloud NAT, firewall, and a test VM.

Implementation

- Create a Cloud Router and Public Cloud NAT for the private subnet.
- Deploy a private VM with no external IP and a dedicated service account.
- Allow only required egress and administration paths.
- Configure OS Login and access the VM through IAP TCP forwarding.
- Enable relevant firewall, flow, and NAT logging for the lab.
- Prove outbound access, denied unsolicited inbound access, and a denied nonrequired egress path.
- Use Connectivity Tests or equivalent evidence to document the route.

Validation Evidence

- The VM reaches approved external and Google API destinations without a public IP.
- Direct public inbound access is impossible.
- Administrative access is identity-controlled and audited.
- Logs identify NAT translation and firewall decisions.

Cleanup

- Delete the VM, NAT gateway, router, temporary rules, and logs sinks created only for the lab.

Interview Questions and Model Answers

1. Does Cloud NAT allow inbound connections?

No. It provides address translation for supported outbound flows and does not create unsolicited inbound reachability.

2. What is Cloud Router?

A managed BGP control-plane service that exchanges dynamic routes; it does not forward data packets.

3. Why use hierarchical firewall policies?

To enforce central security guardrails across folders or the organization while allowing delegated workload rules.

4. What evidence should support network troubleshooting?

Effective routes and firewall policies, Connectivity Tests, flow/firewall/NAT logs, DNS results, and BGP state.

Chapter Summary

- Network security is layered across hierarchy, VPC, edge, routes, and identity.
- Cloud NAT enables private egress but is not a firewall.
- Cloud Router exchanges routes; it is not a packet-forwarding appliance.
- Secure administration should avoid public management exposure.

Chapter 15 - Cloud Load Balancing, Cloud CDN, and Cloud Armor

Delivering healthy applications globally with managed edge controls

Learning Objectives

- Choose among Application, proxy Network, and passthrough Network Load Balancers.
- Explain forwarding rules, target proxies, URL maps, backend services, health checks, and NEG's.
- Use Cloud CDN and Cloud Armor safely.
- Design health-aware releases and multi-region failover.

15.1 Google Cloud Load-Balancer Families

Google Cloud offers managed load balancers for HTTP(S), TCP, SSL, UDP, and other traffic patterns. Selection depends on protocol, external or internal exposure, proxy versus passthrough behavior, global or regional deployment, client IP requirements, and backend type.

Family	Typical Use	Important Characteristics
Application Load Balancer	HTTP/HTTPS APIs and web applications	Layer 7 routing, URL maps, TLS, Cloud CDN/Armor on supported modes.
Proxy Network Load Balancer	Proxied TCP or SSL traffic	Connection proxy, global or regional choices.
Passthrough Network Load Balancer	TCP/UDP with source IP visibility	Direct server return/path behavior and regional scope by type.
Internal load balancer	Private application or service tiers	Private VIP and regional/cross-region options by product.

15.2 Resource Model and Backends

A forwarding rule and IP address form the frontend. Application Load Balancers also use target proxies and URL maps. Backend services define health checks, session affinity, capacity, timeout, security, and the backends that receive traffic. Backends can include instance groups, zonal or serverless network endpoint groups, Cloud Storage backend buckets, and hybrid or internet endpoints where supported.

15.3 Health, Capacity, and Failure Domains

Health checks must reflect readiness. Backend capacity settings influence traffic distribution and autoscaling. Multi-zone and multi-region backends reduce failure impact, but only if quotas, data dependencies, DNS, certificates, and rollback are also prepared.

- Use dedicated health endpoints that test essential dependencies without causing cascading failure.
- Keep firewall rules for health-check and proxy traffic accurate.
- Monitor backend health, request latency, response codes, saturation, and failover state.
- Test one-zone and one-region failure rather than assuming automatic recovery.

15.4 Cloud CDN and Cache Safety

Cloud CDN caches eligible content at Google edge locations when integrated with supported external Application Load Balancers. Cache keys, TTLs, signed URLs/cookies, invalidation, and origin headers determine correctness and cost. Never cache personalized or sensitive responses accidentally.

Decision	Benefit	Risk Control
Enable caching for static content	Lower latency and origin load	Set explicit cache-control and cache key.
Cache API responses	Can reduce expensive repeated work	Vary by identity-sensitive headers or do not cache.
Invalidate content	Fast correction	Use versioned object names to reduce invalidation dependence.
Signed delivery	Protect private content	Rotate signing keys and validate expiration.

15.5 Cloud Armor and Edge Release Controls

Cloud Armor security policies protect supported load-balanced applications with DDoS defense, IP/geo/rate rules, and web application firewall capabilities. Preview rules, logging, and staged rollout reduce false positives. Weighted or separate backends can support canary and blue/green releases, while certificates and DNS changes require their own rollback plan.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
Backend remains unhealthy	Health path, port, firewall, startup, or named-port mismatch	Test endpoint directly, inspect health-check logs, firewall source ranges, named ports, and backend service config.
Users receive 502/503 responses	No healthy capacity, backend timeout, or application failure	Inspect backend health, logs, timeout, connection limits, and autoscaling state.
Cloud CDN serves stale/private content	Incorrect cache-control, cache key, or invalidation	Inspect response headers, cache status, identity variation, TTL, and object versioning.
Cloud Armor blocks legitimate traffic	Rule priority or expression too broad	Use preview mode, request logs, sampled traffic, narrower conditions, and staged enforcement.

Hands-on Lab 15: Deploy a Protected Global Web Entry Point

Prerequisites

- A regional MIG or other supported backend, DNS optional, and permissions for load balancing, certificates, CDN, and Cloud Armor.

Implementation

- Create a health check and backend service for a multi-zone backend.
- Create a global external Application Load Balancer with a reserved IP and HTTPS configuration.
- Add host/path routing and validate healthy backend registration.
- Enable Cloud CDN for a static cacheable path and verify cache headers/status.
- Create a Cloud Armor policy with one preview rule and one safe enforced rule.
- Generate normal and test traffic, observe logs, and document failover behavior.

7. Perform a small backend canary or simulated blue/green switch and record rollback.

Validation Evidence

- HTTPS is served from the intended frontend.
- Unhealthy instances do not receive traffic.
- Cacheable and noncacheable paths behave correctly.
- Cloud Armor preview/enforcement evidence is available.
- Rollback restores the previous backend without DNS change.

Cleanup

- Delete forwarding rules, proxies, URL maps, backend services, certificates, IP addresses, and policies created only for the lab.

Interview Questions and Model Answers

1. What is the frontend of a Google Cloud load balancer?

A forwarding rule associated with an IP address and protocol/port configuration.

2. Why use a network endpoint group?

To represent endpoints beyond a traditional instance group, including serverless, zonal, internet, or hybrid endpoints depending on type.

3. What is the main Cloud CDN security risk?

Caching personalized, private, or stale content because cache keys and headers are incorrect.

4. How should Cloud Armor rules be introduced?

With logging, preview mode, representative traffic, narrow expressions, and rollback before enforcement.

Chapter Summary

- Choose load balancers by protocol, exposure, proxy behavior, and scope.
- Backend health and capacity drive availability.
- Cloud CDN requires deliberate cache correctness.
- Cloud Armor and release controls should be staged and observable.

Chapter 16 - Cloud DNS, Service Discovery, and Private Service Access

Operating authoritative, private, hybrid, and service-level naming safely

Learning Objectives

- Design public, private, forwarding, and peering DNS zones.
- Use DNSSEC, routing policies, logging, and response policies appropriately.
- Implement hybrid DNS forwarding and split-horizon naming.
- Differentiate DNS discovery from Private Service Connect and managed-service access.

16.1 Cloud DNS Zone Types

Cloud DNS provides public authoritative zones and private zones associated with VPC networks. Forwarding zones send selected suffixes to external DNS servers. Peering zones make another VPC network responsible for name resolution. Server policies support inbound and outbound forwarding and DNS logging.

Zone/Policy	Purpose	Primary Risk
Public zone	Internet authoritative DNS	Registrar delegation, DNSSEC, TTL, and accidental record change.
Private zone	Names visible to authorized VPCs	Overlapping suffixes and network authorization.
Forwarding zone	Send suffix to target DNS servers	Routing, target availability, and forwarding loops.
Peering zone	Resolve through another VPC DNS namespace	Non-transitive assumptions and ownership confusion.
Server policy	Inbound/outbound forwarding and logging	Broad scope or unintended override behavior.

16.2 Resolution Order and Split Horizon

Private, forwarding, peering, and public namespaces can overlap. Resolution behavior depends on the zones and policies authorized for the source VPC. Split-horizon DNS intentionally gives different answers internally and externally, but must have clear ownership and testing to prevent accidental leakage or outage.

16.3 DNSSEC, Routing Policies, and Health Checks

DNSSEC protects authenticity for supported public zones when the registrar chain is configured correctly. Routing policies can distribute or steer responses by weighted round robin, geolocation, or failover/health behavior where supported. DNS health is not a substitute for application-level load balancing when rapid convergence or connection draining is needed.

- Lower TTL before planned migration, not during the incident.
- Verify DS records before completing DNSSEC activation.
- Use health-aware DNS only with realistic failure and recovery tests.
- Monitor NXDOMAIN, SERVFAIL, forwarding latency, and resolver logs.

16.4 Hybrid DNS

Inbound forwarding lets on-premises resolvers query private Cloud DNS zones through designated entry points. Outbound forwarding sends selected private queries to on-premises or other resolvers. Hybrid DNS depends on VPN/Interconnect routing, firewall rules, DNS server availability, and avoidance of forwarding loops.

```
gcloud dns managed-zones list
gcloud dns policies list
gcloud logging read 'resource.type="dns_query"' --limit=20
```

16.5 Service Discovery and Private Endpoints

Service Directory can register and discover service endpoints for applications. Private Service Connect creates private endpoints or service attachments for supported producer-consumer patterns. DNS usually maps friendly names to these private endpoints, but DNS itself does not create private connectivity or authorization.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
Private name returns public answer	Private zone not authorized or suffix mismatch	Inspect zone visibility, network association, record name, resolver source, and resolution order.
Hybrid DNS times out	Forwarder routing/firewall/availability or loop	Check inbound/outbound policy, VPN/Interconnect route, UDP/TCP 53, target health, and loop prevention.
DNSSEC domain fails after enablement	Incorrect or stale DS record	Compare Cloud DNS DNSSEC keys with registrar DS and roll back carefully using documented procedure.
Private endpoint name resolves but connection fails	DNS works; route, firewall, endpoint IAM, or service acceptance fails	Inspect PSC endpoint/service attachment, consumer accept lists, route, firewall, and backend health.

Hands-on Lab 16: Build a Hybrid-Ready Private DNS Pattern

Prerequisites

- A VPC network and permissions for Cloud DNS. An on-premises resolver may be simulated with a VM.

Implementation

- Create a private managed zone authorized for the application VPC.
- Add records for an internal service and verify resolution from an authorized VM.
- Create a forwarding zone or outbound server policy for a simulated external suffix.
- Enable DNS query logging and capture successful and failed queries.
- Document an inbound DNS forwarding design for on-premises clients.
- Create a private endpoint DNS design for a Private Service Connect service, even if the endpoint is not deployed.
- Test a split-horizon name and record the internal versus external answer.

Validation Evidence

- Authorized networks resolve private records; unauthorized sources do not.

- Forwarded suffixes reach the intended resolver without loops.
- DNS logs show source, query, response, and latency evidence.
- Private endpoint DNS ownership and TTL are documented.

Cleanup

- Delete temporary zones, policies, test records, and resolver VMs after exporting evidence.

Interview Questions and Model Answers

1. What is a Cloud DNS forwarding zone?

A private zone that forwards queries for a DNS suffix to configured target name servers.

2. Does DNSSEC encrypt DNS queries?

No. It validates authenticity and integrity of DNS data; it does not provide query confidentiality.

3. Why is DNS health checking slower than load-balancer health?

Resolver and client caching based on TTL can delay convergence.

4. Does a private DNS record create network reachability?

No. Routing, firewall, endpoint configuration, and service authorization must also be correct.

Chapter Summary

- Cloud DNS supports public, private, forwarding, and peering patterns.
- Hybrid DNS depends on network paths and loop-free forwarding.
- DNSSEC protects authenticity, not confidentiality.
- Naming and private connectivity are related but separate controls.

Chapter 17 - Cloud Storage Engineering

Designing secure, durable, cost-aware object storage and data lifecycles

Learning Objectives

- Choose locations and storage classes based on access, resilience, and compliance.
- Implement uniform bucket-level access, public-access prevention, encryption, and least privilege.
- Use soft delete, versioning, lifecycle, retention policies, and object holds correctly.
- Operate large-scale transfer, events, performance, and recovery workflows.

17.1 Object Storage Mental Model

Cloud Storage stores immutable object generations in buckets. Replacing an object is an atomic operation that creates a new live generation; recovery behavior depends on soft delete, versioning, retention, and lifecycle settings. Buckets have globally unique names, a location, storage class defaults, IAM policy, and protection configuration.

17.2 Location and Storage Class Selection

Choose regional, dual-region, or multi-region locations based on application placement, data residency, recovery, latency, and cost. Storage classes such as Standard, Nearline, Coldline, and Archive reflect expected access frequency and minimum storage durations. Autoclass can manage class transitions for suitable workloads.

Workload	Location/Class Direction	Critical Check
Active application data	Regional or dual-region Standard	Place near compute; design regional failure behavior.
Enterprise shared dataset	Dual-region or multi-region	Residency and data-transfer economics.
Monthly backup	Nearline or Coldline	Minimum duration and retrieval operations.
Long-term archive	Archive	Restore time, retrieval cost, and retention.
Unknown access pattern	Autoclass candidate	Feature compatibility and cost model.

17.3 Access, Encryption, and Public Exposure

Uniform bucket-level access simplifies authorization by using IAM rather than object ACLs. Public access prevention blocks public IAM grants. Use groups and workload identities, not embedded HMAC keys or service-account keys. Google-managed encryption is default; customer-managed keys are used when key ownership and lifecycle requirements justify the operational dependency.

- Separate data administrators, object writers, readers, and retention administrators.
- Use signed URLs for bounded object access instead of public buckets.
- Log data access for sensitive buckets where required.
- Plan KMS key availability, rotation, and deletion protection.

17.4 Recovery and Immutability Controls

Cloud Storage supports soft delete, Object Versioning, lifecycle management, retention policies, Bucket Lock, object holds, and Object Retention Lock. These controls interact. A lifecycle rule cannot delete data before a retention condition permits it. Locking retention can be irreversible and should follow legal, security, and cost review.

Control	Protects Against	Operational Caution
Soft delete	Accidental or malicious deletion for a retention window	Retained data remains billable.
Object Versioning	Overwrite/delete of live generation	Lifecycle old generations to control cost.
Retention policy/Bucket Lock	Deletion before required age	Locking is irreversible; test governance first.
Object hold	Specific object deletion	Track owner and release process.
Lifecycle rule	Cost and expiry automation	Ordering and conditions can delete data unexpectedly.

17.5 Performance, Transfers, and Events

Object naming, parallelism, resumable uploads, request rates, and regional proximity affect performance. Storage Transfer Service, Transfer Appliance, and managed transfer tools address migration. Event notifications can integrate through Pub/Sub or Eventarc patterns, but consumers must handle duplicates, retries, and ordering assumptions.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
Application receives 403	IAM, conditional grant, public prevention, VPC controls, or KMS issue	Inspect principal, bucket IAM, conditions, access logs, key permissions, and perimeter constraints.
Deleted object cannot be found	Soft delete/versioning not enabled or retention window expired	List generations/soft-deleted objects, inspect lifecycle and retention configuration.
Storage cost rises unexpectedly	Old versions, soft-deleted data, minimum-duration charges, or egress	Use billing export and Storage Insights; inspect generations, class, lifecycle, and location.
Upload fails intermittently	Timeout, proxy, checksum, or non-resumable method	Use resumable upload, retry with idempotency, verify checksum, and inspect client/network logs.

Hands-on Lab 17: Create a Secure Recoverable Object Workflow

Prerequisites

- A dedicated project and permission to create a Cloud Storage bucket.

Implementation

1. Create a uniquely named regional or dual-region bucket with uniform bucket-level access and public access prevention.
2. Grant a test workload identity only the object permissions required.
3. Enable and document soft delete; optionally enable versioning for comparison.
4. Upload an object, replace it, delete the live generation, and perform a recovery test.
5. Create lifecycle rules for noncurrent or test data with a safe dry-run review.
6. Add a retention policy without locking it; document the approval required before Bucket Lock.
7. Capture access, recovery, lifecycle, location, and cost evidence.

Validation Evidence

- The bucket is not publicly accessible.
- The test identity can perform only intended object actions.
- A deleted or replaced object is recovered within the configured protection model.
- Lifecycle and retention interactions are documented.

Cleanup

- Delete all test object generations and the bucket only after retention settings permit deletion and evidence is exported.

Interview Questions and Model Answers

1. What does uniform bucket-level access do?

It disables object ACL-based access and uses IAM consistently at bucket/project scope.

2. Why can versioning increase cost?

Noncurrent generations remain stored until lifecycle or manual deletion removes them.

3. What is dangerous about Bucket Lock?

A locked retention policy cannot be reduced or removed, so data and associated costs can persist until expiry.

4. When use a signed URL?

To grant time-bounded access to a specific object without making the bucket public or giving the caller broad IAM.

Chapter Summary

- Cloud Storage design starts with location, access, recovery, and lifecycle.
- Uniform IAM and public-access prevention reduce exposure.
- Soft delete, versioning, and retention solve different risks.
- Immutability controls require legal and cost governance.

Chapter 18 - Persistent Disk, Hyperdisk, Local SSD, and Filestore

Selecting and protecting block, ephemeral, and managed file storage

Learning Objectives

- Compare Persistent Disk, Hyperdisk, Local SSD, and Filestore.
- Design disk performance, availability, encryption, and attachment patterns.
- Use snapshots, snapshot schedules, regional disks, and asynchronous replication appropriately.
- Operate managed NFS storage and restore workflows.

18.1 Storage Semantics and Selection

Block storage presents devices to VMs; file storage presents shared NFS semantics; object storage exposes API-based objects. Persistent Disk remains broadly available, while Hyperdisk provides newer performance and high-availability options for supported machine types. Local SSD is physically attached ephemeral storage. Filestore is a managed NFS service.

Service	Semantics	Best Fit
Persistent Disk	Durable block volume	General VM boot/data disks and broad compatibility.
Hyperdisk	Durable block with provisioned performance options	Performance-sensitive and supported HA workloads.
Local SSD	Ephemeral local block	Scratch, cache, shuffle, and reconstructable high-I/O data.
Filestore	Managed NFS file share	Shared POSIX file access for VMs, GKE, VMware, or on-premises.
Cloud Storage	Object API	Unstructured data, backups, static content, data lakes.

18.2 Performance Engineering

Disk performance depends on disk type, size, provisioned IOPS/throughput where applicable, machine limits, filesystem, queue depth, I/O size, and application concurrency. Measure end-to-end latency and saturation rather than assuming a larger disk automatically fixes every bottleneck.

- Separate boot and high-I/O data where operationally useful.
- Align filesystem and database settings with failure and consistency requirements.
- Monitor disk throughput, IOPS, latency, and VM limits together.
- Benchmark with realistic I/O patterns and warm-up.

18.3 Availability and Replication

Zonal disks are tied to one zone. Regional Persistent Disk and Hyperdisk Balanced High Availability synchronously replicate across two zones in a region for supported configurations. They can support zonal failover, but applications still need fencing, mount, consistency, and recovery automation. Asynchronous disk replication can support cross-region recovery for supported disk types and scenarios.

18.4 Snapshots and Schedules

Standard and archive snapshots provide incremental disk backups. Snapshot schedules automate recurring protection. Instant snapshots provide fast, location-bound restore points for supported scenarios. Application-consistent backups require guest coordination or managed backup capabilities; a crash-consistent disk snapshot is not automatically database-consistent.

```
gcloud compute snapshots list
gcloud compute resource-policies list
gcloud compute disks snapshot DISK_NAME --zone=ZONE --snapshot-names=SNAPSHOT_NAME
```

18.5 Filestore Operations

Filestore offers service tiers with different capacity, performance, availability, and feature characteristics. Enterprise tiers support high availability and GKE multishares in supported modes. Backups are differential/incremental by chain and can restore shares, including cross-region restore patterns. NFS permissions, network access, mount behavior, and backup performance must be tested.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
Disk latency is high	VM limit, disk type/size, queue, filesystem, or application pattern	Correlate VM and disk metrics; benchmark I/O size, queue depth, and throughput.
Regional disk cannot attach after zone failure	Incorrect failover/force-attach process or stale primary	Follow documented fencing and force-attach workflow; verify recovery checkpoint and application consistency.
Snapshot restores but application is corrupt	Snapshot was crash-consistent or dependencies mismatched	Use application quiescing, coordinated backups, transaction logs, and recovery validation.
Filestore mount fails	Network route/firewall, NFS version/options, DNS, or permissions	Test TCP 2049, route, export, mount options, client identity, and service tier constraints.

Hands-on Lab 18: Benchmark and Protect VM Data Storage

Prerequisites

- A test VM, permissions for disks/snapshots, and optional Filestore quota.

Implementation

1. Create separate boot and data disks using an appropriate Persistent Disk or Hyperdisk type.
2. Format and mount the data disk with a persistent device identifier.
3. Run a controlled I/O benchmark and record VM/disk limits and observed metrics.
4. Create a snapshot schedule and take an on-demand snapshot after writing test data.
5. Restore the snapshot to a new disk and validate file checksums.
6. Document a regional-disk or Hyperdisk HA failover design for the workload.
7. Optionally create a small Filestore share, mount it from two clients, and test backup/restore.

Validation Evidence

- Measured IOPS/throughput/latency are compared with configured limits.
- The restored disk contains validated data.
- Snapshot retention and storage location are documented.
- The failover procedure includes fencing and application consistency.

Cleanup

- Delete benchmark VMs, disks, snapshots, schedules, and Filestore resources not intentionally retained.

Interview Questions and Model Answers

1. What is the key risk of Local SSD?

Data is ephemeral and can be lost during VM deletion, certain host events, or unrecoverable failure.

2. Why is a regional disk not a complete HA solution?

It protects block data across zones, but compute recovery, fencing, mounting, application state, and client failover remain customer responsibilities.

3. Are disk snapshots always application-consistent?

No. Without guest or application coordination they are generally crash-consistent.

4. When choose Filestore?

When applications require managed shared NFS/POSIX file semantics rather than block or object storage.

Chapter Summary

- Choose storage by semantics, performance, availability, and recovery.
- Disk performance is an end-to-end system limit.
- Regional replication and snapshots solve different failure modes.
- Restores and failovers must be tested at application level.

Chapter 19 - Hybrid and Multicloud Connectivity

Connecting data centers, branches, and clouds with resilient routing and operations

Learning Objectives

- Select HA VPN, Dedicated/Partner/Cross-Cloud Interconnect, and Network Connectivity Center.
- Explain Cloud Router, BGP, VLAN attachments, redundancy, and encryption choices.
- Design hybrid DNS, route advertisement, MTU, and observability.
- Plan migrations and cutovers with rollback and capacity evidence.

19.1 Connectivity Product Selection

Cloud VPN extends networks through IPsec tunnels. HA VPN provides a highly available gateway design when paired correctly with redundant peer devices and tunnels. Dedicated Interconnect connects directly to Google; Partner Interconnect uses a service provider; Cross-Cloud Interconnect connects supported other cloud environments. Product selection depends on bandwidth, latency, encryption, availability, location, provider, lead time, and cost.

Option	Best Fit	Major Considerations
HA VPN	Fast deployment, encrypted moderate-bandwidth connectivity	Internet path, tunnel throughput, redundant gateways and BGP.
Dedicated Interconnect	High-bandwidth direct enterprise connectivity	Colocation, lead time, physical redundancy, encryption requirement.
Partner Interconnect	Provider-managed access where direct colocation is unsuitable	Provider SLA, VLAN attachments, bandwidth and locations.
Cross-Cloud Interconnect	Direct supported cloud-to-cloud connectivity	Provider availability, routing domains, security, cost.
HA VPN over Interconnect	Encrypted traffic over Interconnect	Additional routing/tunnel complexity and capacity.

19.2 Cloud Router and BGP Design

Cloud Router establishes BGP sessions and exchanges dynamic routes for VPN and Interconnect. High availability requires redundant routers, tunnels or VLAN attachments, and independent failure paths. Use custom advertisements carefully, control route priority, and validate asymmetric routing.

19.3 Network Connectivity Center

Network Connectivity Center (NCC) provides hub-and-spoke orchestration for VPC, hybrid, and supported producer or gateway spokes. It can simplify connectivity among projects, sites, and clouds, but route exchange, data transfer, segmentation, and organizational ownership must still be designed.

- Separate connectivity administration from workload routing changes.
- Document which spoke types exchange routes and under what topology.
- Use organization-level inventory and logs for hubs, spokes, and route events.
- Model data-transfer charges and traffic inspection paths.

19.4 Reliability, MTU, and Encryption

A reliable design uses independent circuits or tunnels, devices, locations, Cloud Routers, and VLAN attachments. MTU mismatch and fragmentation can cause silent application problems. Interconnect traffic is not automatically encrypted at the IP layer in every design; use MACsec where supported or HA VPN over Interconnect when requirements demand encryption.

- Test loss of each path independently.
- Monitor BGP sessions, tunnel health, packet loss, latency, and capacity.
- Set alert thresholds before production cutover.
- Validate jumbo frames only when the complete path supports them.

19.5 Migration and Cutover Engineering

Connectivity migration should use parallel paths, controlled route preference, DNS/TTL preparation, application dependency mapping, throughput tests, and rollback. Large data migration may require Storage Transfer Service or Transfer Appliance rather than relying solely on production network links.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
Tunnel is up but no traffic flows	Routes, firewall, BGP advertisements, overlapping ranges, or asymmetric return	Inspect Cloud Router status, effective routes, firewall logs, peer routes, and packet path.
Interconnect attachment has low throughput	Circuit/VLAN capacity, TCP window, MTU, QoS, or application limit	Measure each layer, test parallel streams, validate MTU and provider capacity.
NCC spokes do not exchange expected routes	Spoke type/topology or route propagation constraints	Inspect hub/spoke state, route tables, VPC routing, hybrid resource association, and documentation.
Cutover causes intermittent failures	Asymmetric routing, DNS caching, stateful firewall, or partial dependency migration	Compare forward/return routes, session state, DNS TTL, dependency inventory, and rollback path.

Hands-on Lab 19: Design and Test a Redundant Hybrid Connection

Prerequisites

- Two simulated networks or a design environment; billable VPN resources require budget approval.

Implementation

1. Document business bandwidth, latency, RTO, encryption, and availability requirements.
2. Choose HA VPN or Interconnect architecture and justify the decision.
3. Create a redundant topology with two paths, Cloud Router/BGP sessions, and non-overlapping prefixes.
4. Define firewall, route advertisement, DNS forwarding, MTU, and monitoring requirements.
5. If deploying HA VPN, validate both tunnels and advertised routes.
6. Fail one path and record convergence, packet loss, and application behavior.
7. Prepare a production cutover and rollback runbook.

Validation Evidence

- No single tunnel/router/device failure removes all connectivity.
- Expected routes are present and unauthorized routes are absent.
- Hybrid DNS and firewall dependencies are documented.
- Failover metrics meet the stated objective or gaps are recorded.

Cleanup

- Delete billable VPN gateways, addresses, routers, and test VMs when the lab is complete.

Interview Questions and Model Answers

1. What is Cloud Router?

A managed BGP control-plane service used with VPN and Interconnect to exchange dynamic routes.

2. When choose Dedicated Interconnect over HA VPN?

When high predictable bandwidth, lower latency, and direct connectivity justify colocation, lead time, and redundancy cost.

3. What does Network Connectivity Center provide?

Central hub-and-spoke orchestration for supported VPC and hybrid connectivity resources.

4. Why test path failure before cutover?

A diagram does not prove BGP convergence, route symmetry, capacity, DNS, or application recovery.

Chapter Summary

- Hybrid connectivity requires redundant paths and routing control.
- Cloud Router and BGP determine dynamic reachability.
- NCC simplifies orchestration but does not eliminate topology design.
- Cutovers require capacity, dependency, and rollback evidence.

Chapter 20 - Backup, Disaster Recovery, and Core Infrastructure Capstone

Engineering recoverability from business objectives through tested restoration

Learning Objectives

- Differentiate availability, redundancy, backup, replication, and disaster recovery.
- Translate business impact into RTO, RPO, retention, and recovery tiers.
- Use snapshots, Backup and DR Service, backup vaults, and regional architecture appropriately.
- Execute a core-infrastructure recovery game day with evidence.

20.1 Availability Is Not Backup

A multi-zone application can remain available during infrastructure failure, but it can still replicate accidental deletion, corruption, or malicious change. Backup preserves recoverable history. Replication reduces data loss or recovery time for selected failures. Disaster recovery coordinates people, infrastructure, data, identity, DNS, dependencies, and communications after a major event.

Capability	Primary Goal	Does Not Automatically Provide
Multi-zone availability	Continue service during zonal failure	Historical recovery from deletion or corruption.
Replication	Maintain another current copy	Independent immutable history or correct application state.
Snapshot/backup	Recover prior data or resource state	Running application, networking, DNS, or tested orchestration.
Disaster recovery	Restore business service after major interruption	Zero cost, zero data loss, or automatic success without tests.

20.2 RTO, RPO, and Recovery Tiers

Recovery time objective (RTO) is the maximum acceptable outage duration. Recovery point objective (RPO) is the maximum acceptable data loss measured in time. Smaller objectives usually require more replication, automation, reserved capacity, and testing. Classify applications into tiers and define recovery ownership, dependencies, and evidence.

Tier	Example Target	Architecture Direction
Critical	Minutes RTO; near-zero/minutes RPO	Multi-zone/region, replicated data, automated failover, frequent game days.
Important	Hours RTO; hourly RPO	Warm recovery capacity, scheduled backups, scripted rebuild.
Standard	One day RTO/RPO	Backup and restore with documented infrastructure recreation.
Archive	Days	Low-cost retained data with tested retrieval process.

20.3 Google Cloud Data-Protection Building Blocks

Disk snapshot schedules automate incremental protection for Persistent Disk and Hyperdisk. Regional disks and Hyperdisk HA provide synchronous zonal replication for supported workloads. Asynchronous disk replication supports selected cross-region scenarios. Cloud Storage offers soft delete, versioning, lifecycle, retention, and location choices. Filestore provides backups and restore. Managed databases have service-specific backup and replication controls covered in Part C.

20.4 Backup and DR Service

Backup and DR Service provides centralized, policy-based backup management, monitoring, reporting, and recovery for supported workloads. Backup vaults can provide immutable, indelible backup storage to improve cyber resilience. Compute Engine backup plans can protect instance data, and application-consistent capabilities can coordinate pre/post scripts for supported workflows. Always verify current support, frequency, location, and restore constraints.

- Separate backup administration from workload administration.
- Protect backup projects, vaults, service accounts, and KMS dependencies.
- Monitor job success, age of last recovery point, retention compliance, and restore test outcomes.
- Do not treat a successful backup job as proof of recoverability.

20.5 Disaster-Recovery Architecture Patterns

Patterns include backup-and-restore, pilot light, warm standby, and active-active. Choose based on RTO/RPO, data consistency, cost, operations maturity, and regulatory location. Infrastructure as Code, versioned images, replicated artifacts, DNS/load-balancer readiness, quotas, and identity dependencies determine whether recovery is repeatable.

20.6 Capstone Architecture

The capstone is a private, multi-zone Compute Engine service behind a protected global Application Load Balancer. A regional MIG supplies replaceable capacity. Cloud Storage stores protected application artifacts and logs. Persistent Disk or a managed data layer has scheduled protection. Shared VPC-style controls, Cloud NAT, private administration, DNS, monitoring, and a documented regional recovery path complete the design.

```

Users -> Global Application Load Balancer + Cloud Armor + CDN
      |
      Regional MIG across zones
      |
      Private subnet -> Cloud NAT / Google APIs
      |
      Protected disk/data + Cloud Storage recovery copies
      |
      Backup vault / snapshots / alternate-region runbook
  
```

20.7 Recovery Game Day

A game day declares a scenario, freezes unrelated changes, captures baseline health, executes failure or restore steps, measures RTO/RPO, validates data and security, and records gaps. The exercise should include one failed restore or missing dependency to test diagnosis and decision-making, not only the happy path.

Troubleshooting Matrix

Symptom	Likely Cause	Evidence and Corrective Action
Backup job succeeds but restore fails	Permissions, KMS, image, dependency, capacity, or application consistency gap	Perform isolated restore tests and validate boot, data, identity, DNS, network, and application checks.
Recovery region cannot launch capacity	Quota, machine availability, reservation, or image/artifact missing	Prevalidate quotas and machine types; replicate artifacts and maintain alternatives.
Failover serves stale or split data	Replication lag, DNS caching, dual writers, or weak fencing	Measure recovery point, enforce writer fencing, control TTL, and reconcile transactions.
Backup vault cannot be managed during incident	Privilege separation, identity dependency, or perimeter issue	Maintain tested break-glass procedures, workforce access, audit, and protected recovery admin path.

Hands-on Lab 20: Part B Capstone - Recover a Protected Multi-Zone Service

Prerequisites

- Completion of Chapters 11-19, a dedicated project, and budget approval.

Implementation

- Create versioned instance templates and a regional MIG distributed across zones.
- Place the fleet in a private subnet with Cloud NAT, IAP administration, minimal firewall rules, and a dedicated service account.
- Publish the service through a global external Application Load Balancer with health checks, Cloud Armor, and optional CDN.
- Create private and public DNS records as appropriate and document TTL/rollback.
- Store artifacts in a protected Cloud Storage bucket and protect VM data through snapshots or Backup and DR Service.
- Capture baseline health, cost, quota, route, identity, and recovery evidence.
- Simulate an instance and zonal failure; measure service impact and repair.
- Restore protected data to an isolated recovery VM and validate checksums/application behavior.
- Execute an alternate-region recovery tabletop or deployment, measure RTO/RPO, and document gaps.
- Perform cleanup and prove no orphaned load-balancer, disk, snapshot, IP, NAT, or backup resources remain.

Validation Evidence

- The service survives an instance failure and maintains multi-zone capacity.
- Only approved network and identity paths work.
- A protected data copy is restored and validated independently.
- Measured RTO/RPO are compared with business targets.
- The final evidence package includes architecture, commands, logs, costs, gaps, and remediation owners.

Cleanup

- Delete all capstone resources not intentionally retained. Confirm retention locks before deletion and preserve only nonsecret evidence.

Interview Questions and Model Answers

1. What is the difference between RTO and RPO?

RTO is the maximum acceptable outage duration; RPO is the maximum acceptable amount of data loss measured in time.

2. Why is replication not backup?

Replication can immediately copy corruption or deletion and often lacks independent historical recovery points.

3. What proves a backup design works?

A tested restore that validates data, application function, identity, networking, dependencies, timing, and security.

4. What should a DR game day measure?

Actual RTO/RPO, data correctness, failover/failback behavior, communication, manual steps, capacity, and discovered gaps.

Chapter Summary

- Availability, backup, replication, and DR solve different risks.
- Business RTO/RPO objectives drive architecture and cost.
- Backup and DR Service centralizes policy and recovery for supported workloads.
- Only tested restoration and game-day evidence prove recoverability.

Appendix A - Google Cloud Core Infrastructure Command Reference

Purpose	Command
List VMs	gcloud compute instances list
Describe VM	gcloud compute instances describe VM --zone=ZONE
Serial output	gcloud compute instances get-serial-port-output VM --zone=ZONE
List templates	gcloud compute instance-templates list
List MIGs	gcloud compute instance-groups managed list
Describe MIG	gcloud compute instance-groups managed describe MIG --region=REGION
List health checks	gcloud compute health-checks list
List networks/subnets	gcloud compute networks list; gcloud compute networks subnets list
List routes	gcloud compute routes list
List firewall rules	gcloud compute firewall-rules list
List firewall policies	gcloud compute firewall-policies list
List routers/NAT	gcloud compute routers list; gcloud compute routers nats list --router=ROUTER --region=REGION
Connectivity tests	gcloud network-management connectivity-tests list
List load-balancer resources	gcloud compute forwarding-rules list; gcloud compute backend-services list
List DNS zones/policies	gcloud dns managed-zones list; gcloud dns policies list
List buckets	gcloud storage buckets list
Describe bucket	gcloud storage buckets describe gs://BUCKET
List disks/snapshots	gcloud compute disks list; gcloud compute snapshots list
List VPN/Interconnect	gcloud compute vpn-tunnels list; gcloud compute interconnects list
Cloud Router status	gcloud compute routers get-status ROUTER --region=REGION
Backup plans	gcloud backup-dr backup-plans list --location=LOCATION

Review current command help and official documentation before destructive or production changes. Product commands and required flags can change. Use explicit project, region, and zone flags in automation.

Appendix B - Cross-Layer Troubleshooting Matrix

Layer	First Evidence	Common Causes
Context	gcloud auth list; gcloud config list	Wrong principal, project, region, zone, or configuration.

Layer	First Evidence	Common Causes
Compute	Instance operations, serial output, guest logs	Image, startup script, quota, service account, disk, or host event.
MIG	Group actions, health states, autoscaler logs	Bad health check, initialization, capacity, template, or update policy.
Routing	Effective routes, Cloud Router status, Connectivity Tests	Wrong priority, missing advertisement, overlap, non-transitive topology.
Firewall	Effective firewall policies and firewall logs	Hierarchy deny, target mismatch, source range, protocol/port, missing return path.
NAT	Cloud NAT config/logs and source-port usage	Subnet not covered, port exhaustion, route/firewall/DNS issue.
Load balancing	Backend health, forwarding rule, URL map, logs	Health/firewall, no capacity, certificate, timeout, routing rule.
DNS	dig results, DNS policy/zone, query logs	Wrong visibility, delegation, forwarding loop, TTL, DNSSEC.
Storage	IAM/data-access logs, generations, lifecycle/retention	Permission, key, soft delete/versioning, lifecycle, location, minimum duration.
Hybrid	Tunnel/attachment, BGP state, routes, packet loss	Peer config, redundancy, ASN, MTU, asymmetric routing, capacity.
Recovery	Backup job, restore logs, quotas, dependencies	Unrestored identity/network/key, inconsistent data, capacity, untested runbook.

Appendix C - Core Infrastructure Production-Readiness Checklist

- ☐ VM fleets use versioned templates/images, dedicated service accounts, private administration, labels, logs, and documented maintenance behavior.
- ☐ Regional MIGs, health checks, autohealing, autoscaling, canary deployment, and rollback have been load-tested.
- ☐ Shared VPC roles, subnet delegation, IP allocation, route ownership, and service-project boundaries are documented.
- ☐ Hierarchical and VPC firewall rules implement least privilege with controlled exceptions and logging.
- ☐ Cloud NAT capacity, routes, DNS, egress policy, and port utilization are monitored.
- ☐ Load-balancer family, frontend scope, backend health, certificates, CDN cache rules, and Cloud Armor policies are validated.
- ☐ Public, private, forwarding, and hybrid DNS ownership, TTL, DNSSEC, logging, and rollback are defined.
- ☐ Cloud Storage location, access, encryption, soft delete/versioning, lifecycle, retention, and public prevention are approved.
- ☐ Disk/Hyperdisk performance, replication, snapshots, application consistency, and restore tests meet workload objectives.
- ☐ Filestore service tier, NFS access, backup, restore, capacity, and performance are tested where used.
- ☐ VPN/Interconnect/NCC paths have independent redundancy, route controls, MTU validation, monitoring, and failure tests.
- ☐ RTO/RPO, recovery tiers, quotas, artifacts, images, identity dependencies, KMS, DNS, and alternate-region capacity are documented.

- ☐ Backup administration is separated, backup vault protection is governed, and successful isolated restores are recent.
- ☐ Cost ownership covers static IPs, load balancers, NAT, storage retention, snapshots, backup vaults, and network transfer.
- ☐ Every temporary lab or incident resource has an owner, expiry, cleanup evidence, and no retained secrets.

Appendix D - Glossary

Term	Definition
Application Load Balancer	Layer 7 managed load balancer for HTTP and HTTPS applications.
Backup vault	Managed storage boundary used by Backup and DR Service to protect backup data, including immutable options.
Cloud Armor	Edge security policies for supported Google Cloud load balancers.
Cloud CDN	Google edge caching integrated with supported external Application Load Balancers.
Cloud NAT	Managed network address translation for supported public or private connectivity scenarios.
Cloud Router	Managed BGP control plane for VPN and Interconnect dynamic routing.
Forwarding rule	Load-balancer frontend resource that associates IP, protocol, ports, and target/backend behavior.
Hyperdisk	Google Cloud block storage family with provisioned-performance and HA options for supported compute.
IAP TCP forwarding	Identity-Aware Proxy tunnel for controlled TCP access to private resources.
Managed instance group (MIG)	Template-driven group of VMs with repair, scaling, and update capabilities.
Network Connectivity Center	Hub-and-spoke orchestration for supported VPC and hybrid connectivity resources.
Network endpoint group (NEG)	A collection of endpoints used by load balancing and related services.
Private Google Access	Subnet capability enabling private resources to reach supported Google APIs without external IPs.
Private Service Connect	Private endpoint and published-service framework for consumer-producer connectivity.
Regional disk	Synchronously replicated block volume across two zones in a region for supported types.
Shared VPC	A host-project network shared with attached service projects.
Soft delete	Cloud Storage protection that retains deleted objects for a configured period.

Term	Definition
Stateful MIG	Managed instance group that preserves selected per-instance state through recreation.
RPO	Recovery point objective: maximum acceptable data loss measured in time.
RTO	Recovery time objective: maximum acceptable outage duration.

Official Google Cloud Documentation Index

Topic	Official URL
Managed instance groups	https://docs.cloud.google.com/compute/docs/instance-groups
MIG rolling updates	https://docs.cloud.google.com/compute/docs/instance-groups/rolling-out-updates-to-managed-instance-groups
Shared VPC	https://docs.cloud.google.com/vpc/docs/shared-vpc
VPC overview	https://docs.cloud.google.com/vpc/docs/overview
Hierarchical firewall policies	https://docs.cloud.google.com/firewall/docs/firewall-policies
Cloud NAT	https://docs.cloud.google.com/nat/docs
Load Balancing overview	https://docs.cloud.google.com/load-balancing/docs/load-balancing-overview
Choose a load balancer	https://docs.cloud.google.com/load-balancing/docs/choosing-load-balancer
Cloud DNS overview	https://docs.cloud.google.com/dns/docs/overview
Cloud DNS best practices	https://docs.cloud.google.com/dns/docs/best-practices
Cloud Storage overview	https://docs.cloud.google.com/storage/docs/introduction
Object lifecycle management	https://docs.cloud.google.com/storage/docs/lifecycle
Bucket Lock	https://docs.cloud.google.com/storage/docs/bucket-lock
Persistent Disk	https://docs.cloud.google.com/compute/docs/disks/persistent-disks
Disk data protection	https://docs.cloud.google.com/compute/docs/disks/data-protection
Filestore overview	https://docs.cloud.google.com/filestore/docs/overview
Cloud VPN	https://docs.cloud.google.com/network-connectivity/docs/vpn/concepts/overview
Cloud Interconnect	https://docs.cloud.google.com/network-connectivity/docs/interconnect
Network Connectivity Center	https://docs.cloud.google.com/network-connectivity/docs/network-connectivity-center/concepts/overview

Topic	Official URL
Backup and DR overview	https://docs.cloud.google.com/backup-disaster-recovery/docs/concepts/backup-dr
Disaster recovery planning	https://docs.cloud.google.com/architecture/dr-scenarios-planning-guide

Google Cloud product names, features, regional availability, quotas, pricing, and command syntax evolve. Validate the current official documentation and organization policies before production implementation.

Completion Statement

Volume 8B is complete when the learner can operate a secure Compute Engine fleet, deploy and roll back a regional managed instance group, explain Shared VPC governance, troubleshoot private network paths, publish and protect an application through Google Cloud load balancing, design public/private/hybrid DNS, protect object/block/file data, select hybrid connectivity, and prove recovery through a measured restore and game day.